

Silver-Mordenite for Radiologic Gas Capture from Complex Streams: □ Dual Catalytic CH₃I Decomposition and I Confinement

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Highlights (3-5 needed)

- CH₃I is catalytically cleaved over Ag^o-Mordenite zeolite
- Surface Methoxy Species (SMS) are formed at the Mordenite acid sites
- Iodine is captured by the Ag^o to form pore occluded AgI
- NO is co-adsorbed by the Mordenite
- SMS combine to form methanol, dimethyl ether and methyl nitrates which are captured downstream in addition to excess iodine species (HI, I₂)

Abstract (250 words)

The effective capture and storage of radiological iodine (^{129}I) remains a strong concern for safe nuclear waste storage and safe nuclear energy. Silver-containing mordenite (MOR) is a longstanding benchmark for iodine capture. In nuclear fuel reprocessing scenarios, complex gas streams will be present and the need for high selectivity of all iodine containing compounds is of the utmost importance for safety and the environment. In particular, a molecular level understanding of the sorption of organic iodine compounds is not well understood. Here we probe the structure and distribution of methyl iodide sorbed by silver-containing MOR using a combination of crystallographic and materials characterization techniques including: infrared spectroscopy, thermogravimetric analysis with mass spectrometry, Micro-X-ray Fluorescence, powder X-ray diffraction analysis, and pair distribution function analysis. The iodine is captured inside the MOR pore in the form of AgI nanoparticles, that is consistent with the pores sizes of the MOR, indicating that the molecule is both physically and chemically captured in the Ag-MOR. The organic component is surface catalyzed by the zeolite via the formation of Surface Methoxy Species (SMS) that result in downstream organics of dimethyl ether and methanol formation.

Keywords.

Mordenite, Methyl Iodide, Synchrotron, Pair Distribution Function (PDF), Fission gas capture

1. Introduction

Safe nuclear energy is inevitably linked to the safe storage of nuclear waste. As such, extreme selectivity and durability are critical to materials used in nuclear separations and long-term waste storage. Numerous radionuclides with varying radiological decay half-lives have been identified as “volatile radionuclides” including ^{129}I , ^3H , ^{14}C , and noble gases (most notably isotopes of Kr and Xe). The effective separation, capture and storage of these radionuclides are areas of active research.[1] ^{129}I is one of the radionuclides that remains of strong interest and concern. While this radionuclide is only found in small concentrations in nuclear effluent, the effective capture and storage of iodine (found in aqueous streams as H^{129}I , $^{129}\text{I}_2$, and/or organic iodides) is critically important to public safety due to its involvement in human metabolic processes and its long half-life ($\sim 10^7$ years).[2] The leading approach to capture radioactive gaseous by-products such as iodine during nuclear fuel reprocessing involves sorption onto silver-containing zeolites.

Indeed, for many decades the silver-containing zeolite mordenite (MOR, Figure 1) has been a benchmark for radiological iodine capture. Until recently, the molecular basis for its performance had remained largely unexplored.[3, 4] We have successfully undertaken structure-property studies into understanding the structure of Ag-MOR,[5] as well as the capture of I_2 gas by a variety of porous crystalline materials (including zeolites and Metal-organic frameworks (MOFs),[6-9] the storage of the iodine gas in durable waste forms[10, 11], and the combined storage/capture of iodate and iodides from caustic aqueous streams[12]. Worldwide, a larger effort for the capture and storage of iodine is underway. Additional capture strategies include I_2 gas with silver-impregnated amorphous silica[13], chalcogel-based aerogels [14, 15], and MOFs for the extraction of I_2 from organic solutions [16]; additional storage strategies include the incorporation of iodine in structurally compatible apatite mineral analogs[17, 18] and iodine immobilization via hot isostatic pressing.[19]

Because of the competitive adsorption encountered by sorption materials in complex off-gas streams, molecular studies are being extended to the interaction of silver-mordenite with other iodine-containing components. In particular, we extend our studies to the sorption of iodine from methyl iodide ($\text{CH}_3\text{-I}$) by Ag-MOR. In addition to its ^{129}I isotope being a radiological by-product of nuclear fuel reprocessing,[20] methyl iodide is

known as a pesticide[21] and a chemical methylating reagent,[22] including in the methanol-to-gasoline (MTG) catalytic process.[23] These processes benefit from the ability to understand the structure-property relationship between $\text{CH}_3\text{-I}$ and its adsorption properties.

To rationally optimize the capture process, the subsequent processing for long term storage and, ultimately, to develop the next generation of iodine capture materials, it is first important to understand the distribution and properties of iodide formed (AgI) and the role of the zeolite support.

Here we present a differential pair distribution function (d-PDF) study of $\text{CH}_3\text{-I}$ -treated silver-containing MOR to probe the structure and distribution of the supported AgI —the organic iodine-capture vehicle. The pair distribution function (PDF) method probes the local structure of nano-scale and amorphous materials, where conventional Bragg crystallographic analysis yields only limited insight. It provides the distribution of atomic distances within a material, and while not intrinsically chemically specific, by using a differential approach to PDF analysis[24] it is possible to isolate contributions from individual components within a material.[25] This involves subtracting a reference PDF, measured for the host zeolite, from the iodine-loaded system, such that atomic distances from the zeolite can be separated from those in the supported material. It is a standard practice to reduce the Ag^+ -exchanged MOR prior to use. Here, the capture of $\text{CH}_3\text{-I}$ from a complex acidic stream by a reduced silver MOR is investigated.

Significant complexity of the organic iodide (guest) loaded Ag-MOR (host) results from the inherent challenge in decoupling the guest structure and framework distortion from the host lattice. Therefore, in order to develop an understanding of the structure–function relationship, a combined experimental approach was necessary. This includes infrared (IR) spectroscopy, thermogravimetric analysis (TGA) with mass spectrometry (MS), Micro-X-ray Fluorescence ($\mu\text{-XRF}$), powder X-ray diffraction (XRD) analysis, and pair distribution function (PDF) analysis.

2. Experimental

2.1 CH₃-I loaded Ag[°]-MOR. Ag[°]-MOR was provided by Oak Ridge National Laboratory (ORNL). IONEX Ag-MOR (AW-300) is a silver exchanged natural mordenite. A sample with reduced silver metal was prepared by heating at 150°C in flowing 3% H₂ for approximately 12 hours, resulting in a material with a standard formula of (Ag,H*)_{1.52}Fe_{0.06}Ca_{0.11}K_{0.16}Al_{2.35}Si_{9.47}O₂₄•4H₂O, H* = charge balancing proton adsorbed from the reducing steam.[26, 27]

Static Adsorption Experiments. A ceramic boat holding Ag[°]-MOR (natural) is placed in a tube furnace. A CH₃-I vapor in a N₂ carrier gas, flowing at ~ 2 ml/min, is streamed over the zeolite for 24 hours at 150°C. Excess nonadsorbed CH₃-I was captured in a water bath outside the tube furnace. This resulted in a loading of 8.5 wt% I on the Ag-MOR. The resultant loaded MOR sample lightened in color from dark gray to yellowish-gray.

Deep Bed Iodine Sorbent Test Experiments. (Figure 2) Approximately 10g of the Ag[°]-MOR was loaded into a vertical sorption bed column (1.9 cm diameter, 3.8 cm length) that was located in a temperature controlled oven. The inlet mixed gas stream approximated that consisted of 1.8% water vapor, 800 ppm NO, 800 ppm NO₂, and 57 ppm CH₃I in a carrier stream of air, with a flow rate of 0.73 L/min and at 150°C.[28] Sorption experiment time was approximately 1 month. Post-sorption samples for structural analysis were taken from 1.25 to 5 cm bed depth.

2.2 High-energy synchrotron X-ray scattering measurements. High energy X-ray scattering data were collected at beamline 11-ID-B at the Advanced Photon Source at Argonne National Laboratory for sample recovered from the online flow sorption experiments, loaded in polyimide capillaries. Total scattering data suitable for PDF analysis were collected at ~87 keV, $\lambda = 0.1430 \text{ \AA}$ for the sample recovered from the sorption experiment and at ~90 keV, $\lambda = 0.1370 \text{ \AA}$ for unreduced Ag⁺-MOR (for the reference PDF) and the pristine / unexposed Ag[°]-MOR sample used for sorption experiment. A large amorphous silicon-based area detector was used to record two-dimensional diffraction images.[29, 30] These data were reduced to one-dimensional scattering data within fit2d.[31]

2.3 Rietveld structural analysis. To model the X-ray diffraction data the atomic coordinates of the framework atoms were taken from previous reports. Refinements are

carried out in GSAS. Initial refinements of the XRD patterns entailed modeling of the atom coordinates for the Al-Si-O framework only, with no attempt to refine site positions and occupancies of the species within the large cavities of the zeolite. These initial refinements on the Al-Si-O cage included refinement of variables for lattice parameters, along with the scale factor, peak shape parameters, and background. Upon completion of this initial fit, Difference-Fourier maps were employed to visualize the unmodeled electron density within the pore cavities. Locations of candidate sites for Ag, water, iodine, and other possible species were established based on scattering magnitude and general coordination to the zeolite framework. Final refinements included site occupancy and positional parameters for the species present in the pore with the site occupancies restrained to be consistent with known elemental composition determination from X-ray Fluorescence (XRF) results. Framework atoms were modeled with a B_{iso} (isotropic temperature) value of $\sim 1.0 \text{ \AA}^2$. Atom species present in the pores were modeled with B_{iso} of $4.0 \text{ \AA}^2$.

2.4 Pair Distribution Function (PDF) analysis. The PDFs, $G(r)$, were obtained from the total scattering data, collected to high values of momentum transfer, $Q_{\text{max}} = 20 \text{ \AA}^{-1}$, using PDFgetX2.[32] Contributions from the background, Compton scattering, fluorescence, to the total scattering data were subtracted as described previously.[29] Differential PDFs (d-PDFs) which isolate the correlations involving the species within the zeolite pores were recovered by subtracting the PDF for the unreduced Ag^+ -MOR as described previously.[6] Structural models were refined against the d-PDFs within PDFgui.[33] The intensities of selected peaks were quantified by fitting Gaussian functions within fityk.[34]

2.5 Thermogravimetric-Differential Scanning Calorimetry (TGA-DSC) and Mass Spectrometry (MS) analyses. The thermal desorption of guests and eventual framework decomposition was investigated by TGA-DSC-MS. Measurements used a SDTQ600 TA instrument, equipped with a mass-spectrometer gas analyzer MS-ThermoStarTM from Pfeiffer Vacuum. Samples were heated at $10^\circ\text{C}/\text{min}$ to 800°C under nitrogen flow.

2.6 Micro-X-ray fluorescence. The μ -XRF analysis was performed using a Bruker M4 Tornado μ -XRF mapping system. The instrument was equipped with a micro-

focused Rh source (50 kV, 600 μ A) with a poly-capillary optic (~ 30 μ m spot-size). Fluorescence spectra were collected using two silicon-drift detectors. Powder specimens were loaded in small Teflon cups (~ 2 mm inner diameter), then secured to the xy-translation stage within the M4, and XRF spectra were collected under vacuum ($\sim 10^{-3}$ Torr). Quantification of the elements (Si, Al, Ag, I, Ca, Mn, and Ca) was possible using the “Quantitative analysis” routines within the M4 software. As the data is uncalibrated, only approximate elemental quantities are available.

2.7 Gas Chromatography – Flame Ionization Detector (GC-FID). Gaseous methyl iodide concentrations are measured using a gas chromatograph (GC; Hewlett Packard 5890 GC) with a flame ionization detector (FID), and a solid-phase micro-extractor (SPME) to improve the GCFID detection limit to about 10 ppb for methyl iodide. Gas samples can be injected into the GC using a sample loop, syringes, or solid-phase micro-extraction (SPME) syringes, which improve the detection limit by approximately 100x.

2.8 Infrared Spectroscopy (IR). Data are collected on a Thermo Scientific Nicolet iS10 Smart iTR. Samples are sub-milligram in size. Each sample data set is collected in air and in the range of 3500 - 500 cm^{-1} .

3. Results

Mordenite, $(\text{Ca}, \text{Na}_2, \text{K}_2)\text{Al}_2\text{Si}_{10}\text{O}_{24} \cdot 7\text{H}_2\text{O}$, is the zeolite-of-choice for radioactive iodine capture due to its high Si:Al ratio which enhances stability in the acidic waste streams and the relatively high Ag-loadings possible. The MOR framework structure defines one-dimensional channels (12-rings, 6.5×7.0 Å) parallel to the crystallographic *c*-axis which contain the charge balancing cations.

Based on weight gained during the deep bed sorption testing, iodine loading onto the Ag° -MOR is approximately 5.2 wt%; this translates into approximately 55% utilization of silver loaded onto the MOR zeolite. However, it is important to note that some of this weight gain may be from adsorbed organic components from interaction of stream gases with the zeolites (e.g., CH_3I , H_2O , or NO_x). Inorganic iodine (either I_2 or HI) is detected downstream of the sorbent beds after column breakthrough occurs, indicating that methyl iodide was being converted into other compounds when in contact with, or through reaction with, the Ag -MOR. Consistent with prior test results,[35, 36] after breakthrough

of the sorbent occurs, a significant portion (from approximately 10 to 90+%, depending on degree of breakthrough) of the iodine that passes through the bed is not methyl iodide, but is either I_2 or HI. The stoichiometric formula calculated from ICP-MS for the Ag° -MOR (natural) sample that is exposed to CH_3I is: $(AgI)_{1.02}(H_{1.44}K_{0.16}Ca_{0.10}Fe_{0.05}Al_{2.53}Si_{9.63}O_{24})$ with residual $Ag = 0.42$. Note, this is a bulk calculation of an inhomogeneous material.

Various organic compounds are detected in the sorption bed gas outlet effluent stream. They include Dimethyl Ether (DME), methyl nitrate, and methanol. These results are generally consistent with similar prior research: Haefner[35] observes methanol in the off-gas and suggests it is a byproduct of methyl iodide adsorption on Ag -MOR, while Scheele[36] proposes that the observed methanol and DME in the off-gas are organic byproducts from methyl iodide adsorption on Ag -MOR.

Sorption onto the Ag° -MOR proceeds at a relatively high level ($\sim 50+$ %) of the Ag utilization and saturation level for the sorbent. When approaching column breakthrough, the sorption process slows, and the outlet effluent gas composition approaches the same composition as the inlet gas, with the addition of the new organic compounds including DME and methanol. A quantitative analysis of the organic byproducts in the sorbent bed outlet gas suggests that not all of the organic moiety (CH_3^+) from the methyl iodide is quantitatively represented by the amounts of the organic byproducts in the outgas effluent gas. This result suggests that some of the organic moiety is also adsorbed on the sorbent, or else is converted into other gaseous byproducts – such compounds as CO , CO_2 , H_2O and/or H_2 .

Refinement of structural models against the d-PDF of Ag° -MOR, prior to CH_3I exposure, indicates that it consists of 8 nm Ag nanoparticles with a face-centered cubic structure ($a = 4.078 \text{ \AA}$, $R \sim 13.4\%$) supported on MOR. (Figure 5) By contrast, the d-PDF for the post-sorption, CH_3I -treated system was dominated by features at short distance, below *ca.* 6 $\text{\AA}$. Refinements of models against the data within PDFgui, suggested that these were predominately subnanometer clusters of an AgI phase (likely α - AgI) with a minor component of larger γ - AgI nanoparticles (11%, *ca.* 6 nm). The largest deviation in the fit of the AgI phases to the data is apparent for the feature(s) at 2.7 - 3.0 $\text{\AA}$ which corresponds to the nearest-neighbor Ag - I distance. While the overall intensity of

this feature is well modeled by the AgI phase, it appears to contain contributions from two distinct peaks with similar distances. These peaks are centered at 2.77 Å and 2.97 Å, with relative peak areas of 3:1. These distinct distances may reflect local defects/distortions in the α -AgI phase.

There was no evidence for residual face-centered cubic metallic Ag⁰, suggesting that either the Ag⁰ has been fully reacted to form AgI or that the Ag⁰ is completely re-oxidized during the on-stream sorption process. Given the distribution of the subnanometer AgI clusters versus larger AgI nanoparticles observed here, and our previous studies that indicate that subnanometer AgI clusters are the more heavily product when iodine (albeit as I₂, not CH₃I) is captured by unreduced Ag⁺-MOR, we propose that a degree of reduction of the supported Ag⁰ metal to Ag⁺ ion likely occurs in-situ/online.

The thermal analysis TGA-DSC-MS data (see Figures 3 and 4) provides a great deal of information about the adsorbed molecules and compounds, plus the stability of the captured iodine. In the TGA, ~0.5 wt% weight loss around 100°C and ~2.5 wt% loss centered around ~150°C is associated with CO₂ loss. In the DSC, an endothermic peak at 152°C is associated with the γ - to α - AgI temperature-induced phase transition in Ag-MOR[6] (the ambient two-phase mixture of β - and γ -polymorphs transforms to the (super) ion-conducting α -phase above 147°C).[37-40] Furthermore, there is a general exothermic slope matching the TGA weight loss seen at ~ 225°C. Clarification of the thermal events is available through the comparison with the coincidental MS data. Below 100°C, there is CO₂ loss that is presumably the combustion product of the excess organic at the bulk surface of the zeolites. Near 160°C, a combination of CO₂, H₂O and possibly -OH are released; this corresponds to occluded and/or pore adsorbed water and organic molecules (the CO₂ is again a product combustion). It is at approximately 225°C that the iodine (I) and CO begin their release, in addition to maximum CO₂ loss. It is important to note that the O₂ and CO rise (release) until the maximum temperature of 600°C, suggesting that they are continuously off gassing due to heating above 225°C. There is no evidence of AgI, NO₂, I₂ loss in the temperature range studied.

As expected, the spot XRF analysis of the CH₃I-Ag-MOR loaded samples detect Ag, Ca and Fe, with Ca and Fe in lower molar fractions than Ag. (Figure S1) This is confirmation of incomplete ion exchange of the silver for the natural mordenite charge

balancing cations. (Figure S2). This data confirms that no AgI is lost to volatilization above its melting point of 556°C, but instead shows the retention of the AgI nanoparticles inside the MOR pores. This information combined with the lack of AgI melting in the TGA/DSC is confirmation that the in-pore confinement of these nanoparticles adds to their stability, and allows for the probably “amorphization”[41] and retention of the AgI at higher temperatures.

Comparison of the as-received Ag^o-MOR and the post-exposure to CH₃I-Ag^o-MOR (see Figure S3) IR spectra reveal no difference in the traces. This is further evidence that there is neither permanent organic functionalization of the zeolite framework, nor is there any capture of CH₃I in the pores of the MOR (as a possible functionalized silver nanoparticle). This data is confirmation of that elucidated from the crystallographic analyses.

4. Discussion

This set of experiments was performed under simulated industrial conditions of fission off-gas capture for nuclear fuel reprocessing. The zeolites are natural and as a result have complex pore cation distributions. Furthermore, they are in bulk pelletized form throughout the study, including both when reduced and when packed into the sorption beds.

The sorption of iodine from CH₃I by Ag-MOR is possible through a series of catalytic and sorption processes. Through bulk elemental analysis of the silver loaded natural mordenite, we observe that the charge balancing cations are Ca, K and presumably H (to compensate for the reduced Ag⁺ to Ag^o in the pores). Therefore it is reasonable to assume that on a localized scale, there were regions of the zeolite that were basic in nature (Ag⁺, Ca²⁺, K⁺) and areas that were acidic in nature (Ag^o and H⁺ in the pores).

Much research has been performed into the catalytic nature of zeolites that are exposed to methyl halides and further the reactivity of surface alkoxy species on zeolites surfaces for the initiation mechanisms of the methanol-to-gasoline (MTG) catalytic process. Their insights are valuable in understanding the methodology of iodine capture of CH₃I by Ag-MOR in complex streams.

In the case of basic zeolites used in the conversion of methyl halides, Haw, et.al.,[42] utilized ^{13}C magic angle spinning (MAS) NMR to study the formation of intermediate surface products on the zeolite including methyl halides converting to surface methoxy group intermediate on route to the formation of ethylene. Furthermore, he showed that depending on the level of basicity of the zeolite (as determined by the charge balancing cations and/or zeolite acidity level), methyl iodide conversion to hydrocarbons reactions commenced at 225°C over CsX zeolite.[43]

In the case of acidic zeolites, Wang, et.al., found similar intermediate species formations (surface methoxy species, SMS; example within Figure 6) via ^{13}C MAS NMR studies. The SMS act as methylating agents on acidic zeolites when they are reacted with a number of molecules including methanol and hydrochlorides. The SMS can then react with an number of molecules in the stream to form various hydrocarbon products. Examples include (1) the SMS reacts with methanol to form dimethyl ether (DME) plus an acidic site on the zeolite; or the SMS reacts with water to form methanol (CH_3OH) plus an acidic site on the zeolite. Wang further showed that SMS on acidic zeolite surfaces produced CO and CO_2 via the oxidation of the SMS in the presence of oxygen.[44]

Based on the data collected on the molecules collected downstream from the CH_3I adsorption over the Ag-MOR, the TGA/DSC-MS of the iodine loaded AgI-MOR and the structural analyses, we postulate the following reactions occur in the process of capturing the iodine from the complex input stream of CH_3I , H_2O , NO, NO_2 and air. First, CH_3I is cleaved to form an iodine species and a surface methoxy species at the low operating temperature of the adsorption process, 150°C . Second, the iodine reacts with the Ag to form AgI nanoparticles inside the zeolite pore. Third, the SMS react with water to form methanol, and then further react with methanol to form dimethyl ether. Both methanol and dimethyl ether are collected downstream of the adsorption column. This is pictorially described in Figure 6.

The TGA-DSC-MS data of the final iodine adsorbed zeolite indicates that at 225°C , there is a release of iodine, NO, CO and CO_2 species from the sample. The NO is probably desorbing from the zeolite pore as they are readily adsorbed by acidified aluminosilicate zeolites, including MOR.[45] CO and CO_2 are possibly produced by the

oxidation of excess and/or unreacted SMS, that are on the zeolite surface, by either sorbed H₂O or nitrates. There is no evidence of AgI loss (melting point 556°C) up to 600°C. Instead, as seen in the literature, it appears that the AgI is forming an “amorphous”[41] phase that continues to reside in the zeolite pores. This has been confirmed by the XRF data that identifies Ag and I in similar ratios in the pre- and post-heated (600°C) CH₃I-Ag-MOR samples.

Interestingly, the resultant AgI-MOR product after CH₃I interactions, appears to be a less stable iodine capture material than that formed from the adsorption of pure I₂ gas on same as-received starting material Ag^o-MOR (natural) to form AgI-MOR (I loss begins at approximately 360°C). (See figure S4) Examination of the data leads to the conclusion that the desorption of NO and CO may be facilitating the loss of excess I (acidified, and in the form of HI) from the pores and bulk surface of the mordenite.

5. Conclusions

The capture mechanism of radiological fission gases is of vital interest to the population and the environment. Through a combination of materials chemistry, complex gas stream deep bed adsorption studies, and crystallographic (X-ray diffraction refinement and Pair Distribution Function analysis) methods, the mechanism of iodine adsorption from methyl iodide in an acidic wet stream is elucidated.

The CH₃I is catalytically cleaved over the partially reduced Ag^o-mordenite natural zeolite. In that cleaving process, surface methoxy species are formed at the mordenite acid sites; the iodine species is either captured by the silver to form occluded AgI in the MOR pores, or it reacts to form unbound I₂ or HI. There is evidence of nitrite species also being adsorbed from the inlet gas stream on the mordenite. As the reaction proceeds, the SMS react to form a number of off-gas species such as methanol, dimethyl ether, and methyl nitrate.

The combination of chemical, thermal and mechanical stability of the zeolites make them strong candidates for the use in nuclear energy applications. The chemical and physical confinement of the radiological species (e.g., Ag¹²⁹I) inside the pores of the zeolite allows for the safe manipulation of the material into a waste form for long-term decay storage. Research into the incorporation of iodine captured from complex streams

into waste forms that are chemically durable (e.g., low temperature sintering Bi-Si oxide glass)[10] for a variety of postulated repository geologies is ongoing.

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Figure Captions

Figure 1. The MOR structure. The MOR framework defines one-dimensional channels (12-rings, 6.5×7.0 Å) parallel to the c -axis which contain the exchangeable cations and water molecules (omitted for clarity);[46] standard formula: $(X^{1+})_2AlSi_{10}O_{24}$.

Figure 2. Iodine sorbent deep bed test system used in the capture of iodine from a complex stream containing CH_3I , H_2O , NO and NO_2 with a carrier gas of air.

Figure 3. Raw PDFs of CH_3I -Ag-MOR (blue) and pristine Ag-MOR (red), which both include the zeolite component. Differential shown in green. PDF shows a standard AgI phase [6] with sub 6 Å particle size: one well defined peak at 2.8Å and a broad feature between 4.5 and 5 Å.

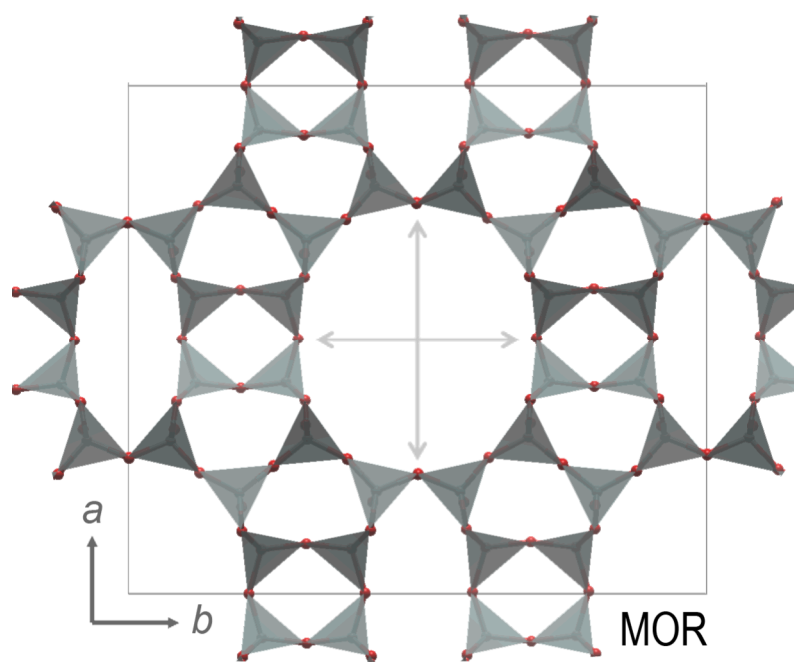
Figure 4. TGA-DSC of CH_3I -Ag-MOR sample in flowing He. Dip at 152°C represents the endothermic, temperature induced, phase transition of the pore occluded γ/β - to α -AgI nanoparticles.

Figure 5. MS data from simultaneous TGA-DSC-MS, run under flowing He.

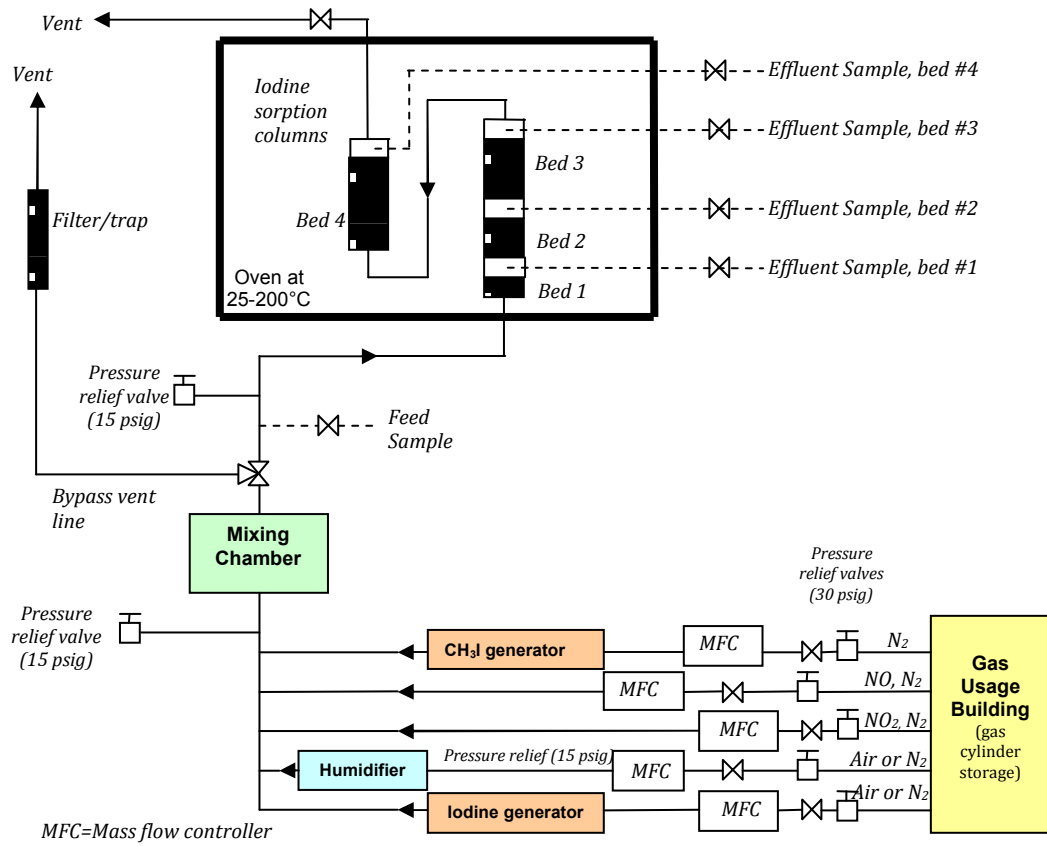
Figure 6. A schematic portraying the dual catalytic CH_3I decomposition and I confinement reaction pathway over the Ag° -MOR zeolite. The *SMS* unit is identified.

Figures

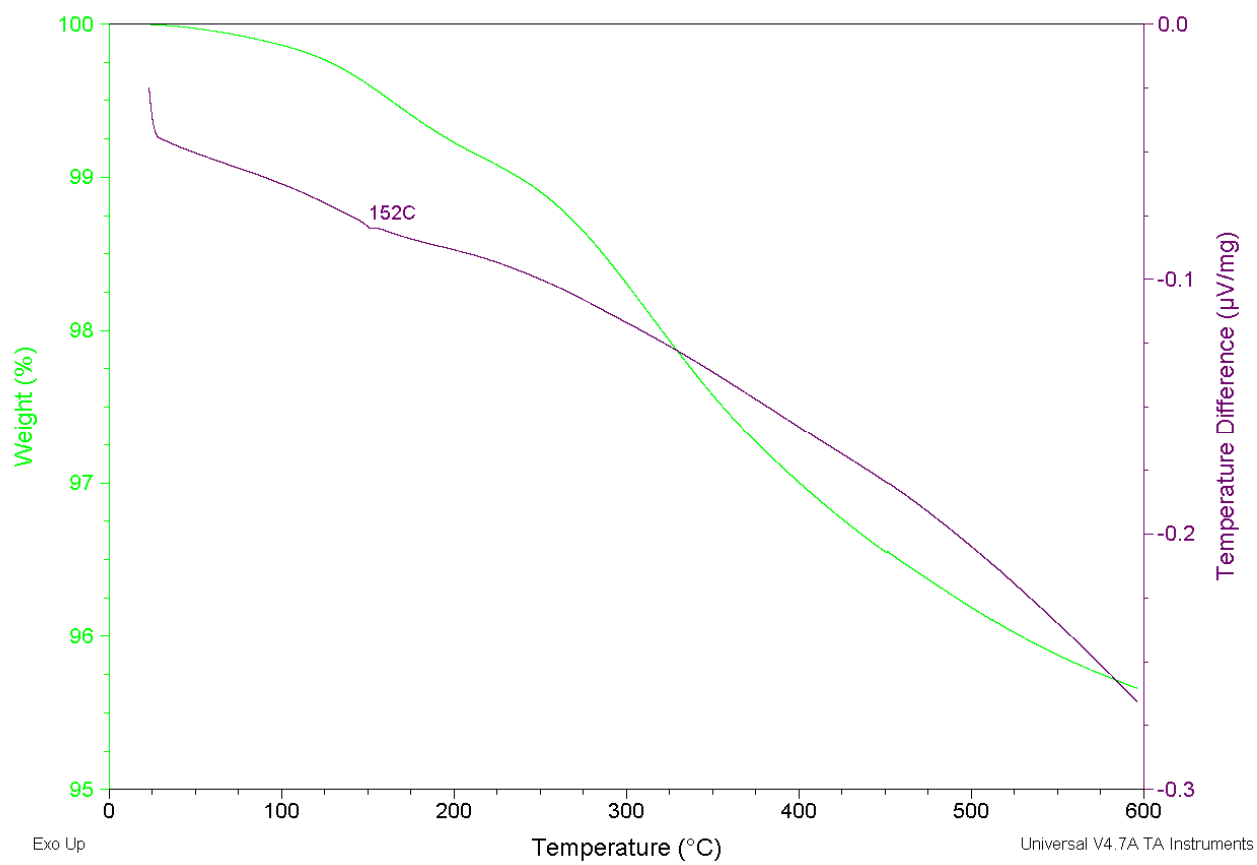
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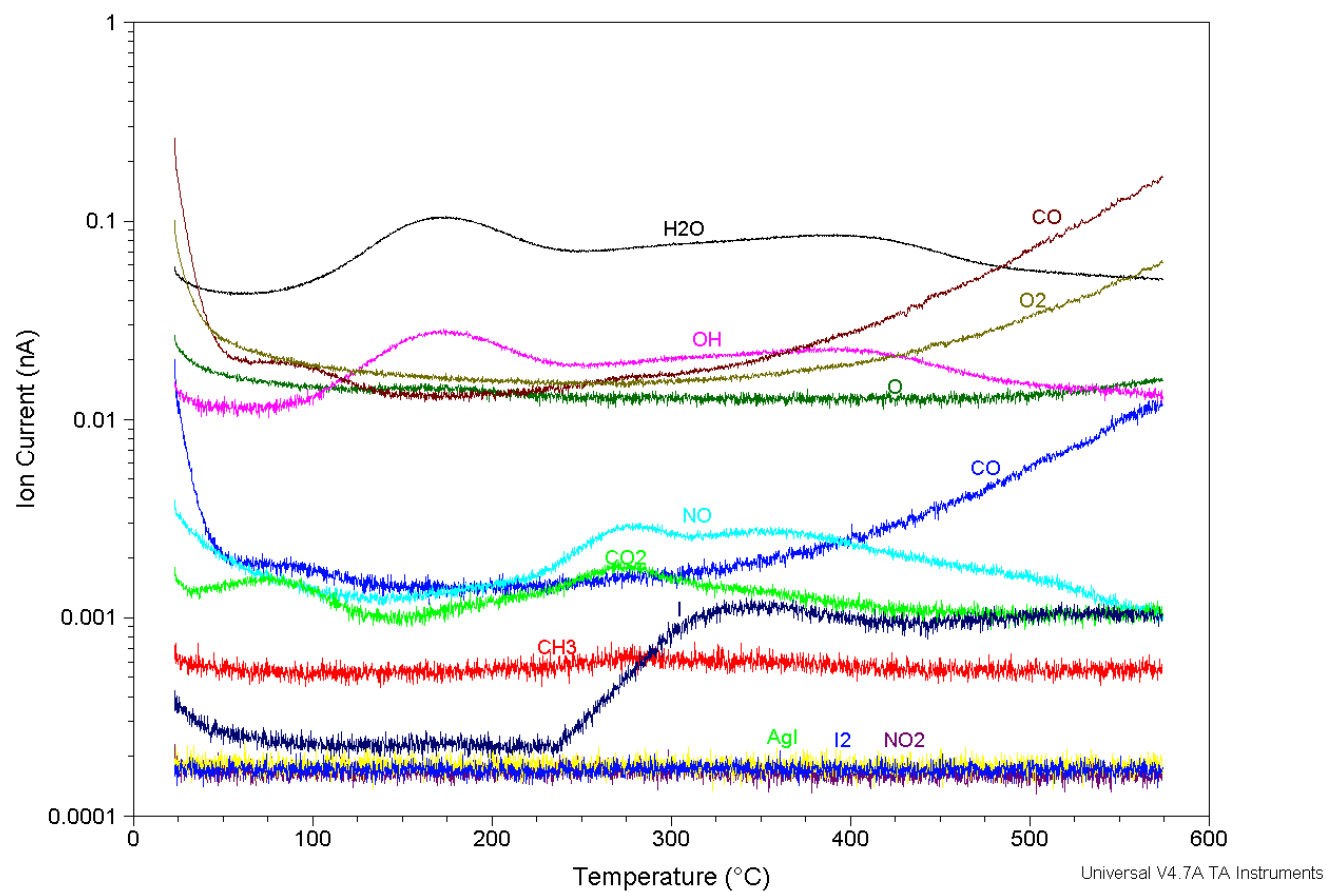
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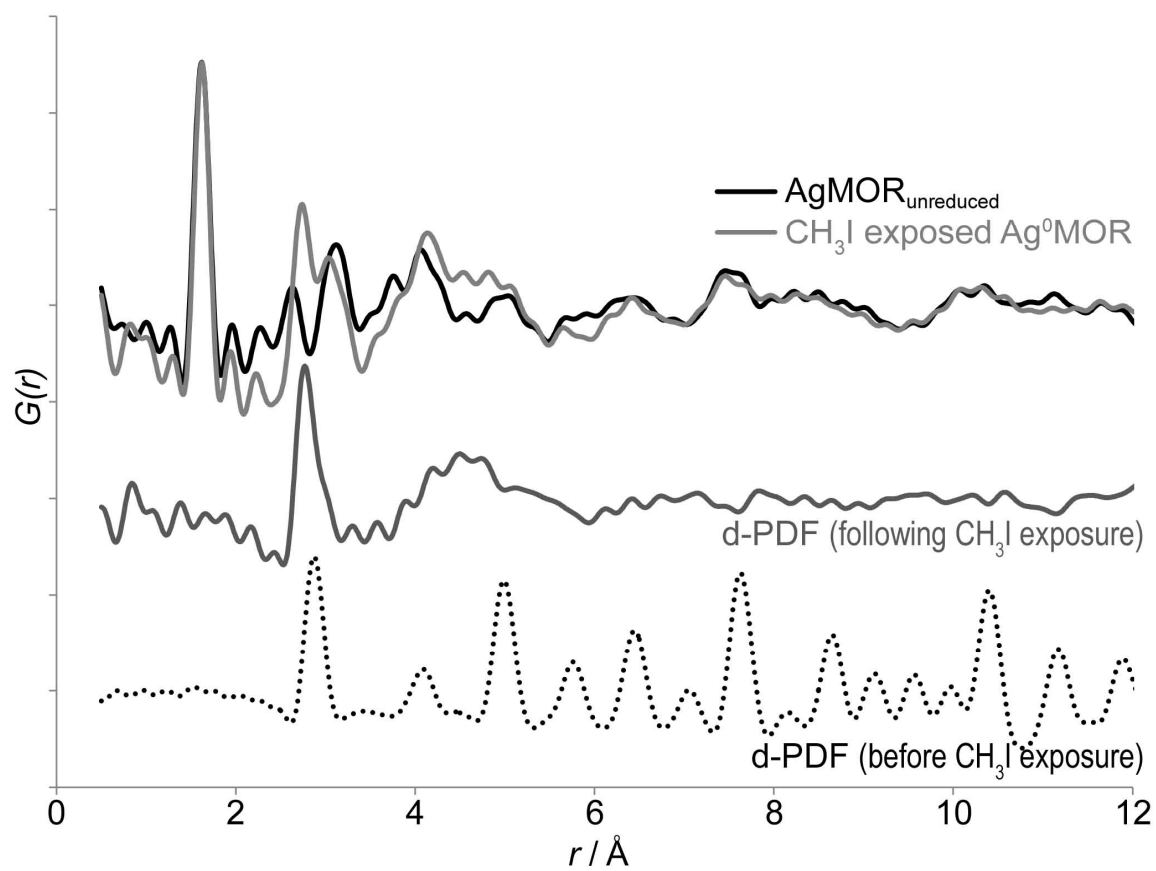
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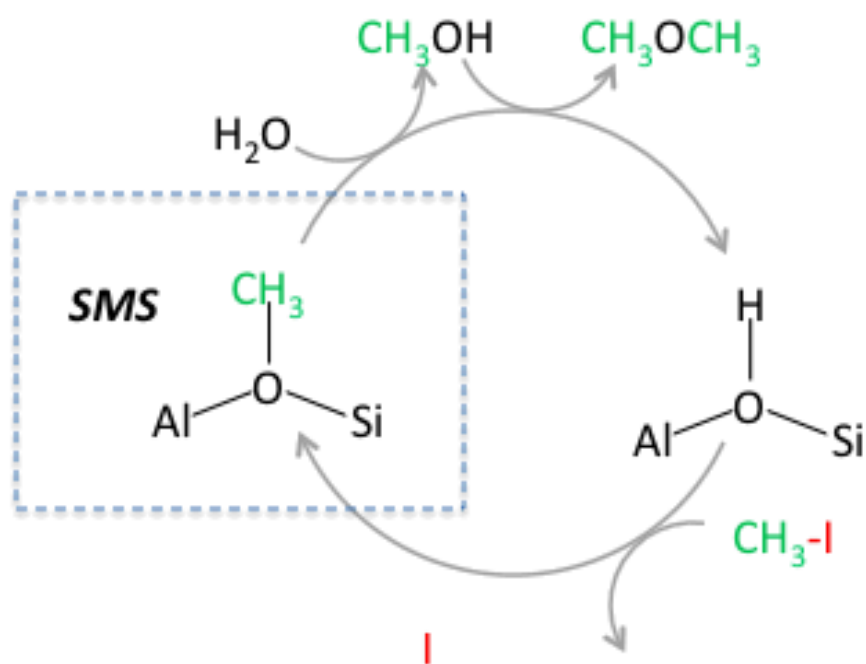
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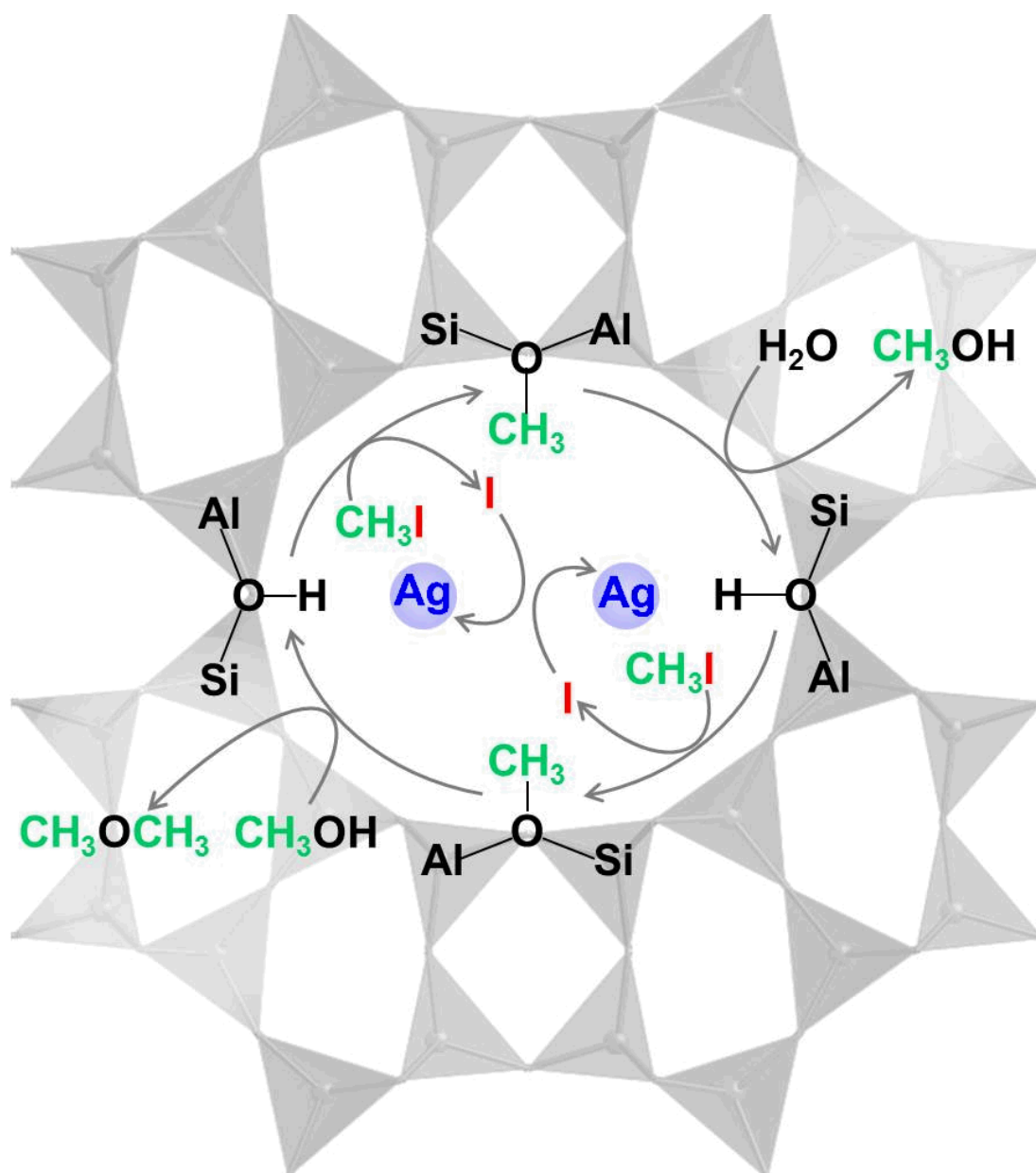
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6.



Graphical Abstract.



References.

- [1] N.R. Soelberg, T.G. Garn, M.R. Greenhalgh, J.D. Law, R. Jubin, D.M. Strachan, P.K. Thallapally, Hindawi: Science and Technology of Nuclear Installations, (2013).
- [2] D.R. Haefner, T.J. Tranter, Report No. INL/EXT-07-12299, Idaho National Laboratory, (2007).
- [3] Treatment, conditioning and disposal of iodine-129, International Atomic Energy Agency, 1987.
- [4] Nuclear Energy Handbook, CRC Press, Taylor and Fancis Group, Boca Raton, FL, 2009.
- [5] H.Y. Zhao, T.M. Nenoff, G. Jennings, P.J. Chupas, K.W. Chapman, J. Phys. Chem. Letters, 2 (2011) 2742-2746.
- [6] K.W. Chapman, P.J. Chupas, T.M. Nenoff, J. Am. Chem. Soc., 132 (2010) 8897-8899.
- [7] J.T. Hughes, D.F. Sava, T.M. Nenoff, A. Navrotsky, J. Am. Chem. Soc., 135 (2013) 16256-16259.
- [8] D.F. Sava, K.W. Chapman, M.A. Rodriguez, J.A. Greathouse, P.S. Crozier, H. Zhao, P.J. Chupas, T.M. Nenoff, Chem. Mater., 25 (2013) 2591-2596.
- [9] D.F. Sava, M.A. Rodriguez, K.W. Chapman, P.J. Chupas, J.A. Greathouse, P.S. Crozier, T.M. Nenoff, J. Am. Chem. Soc., 133 (2011) 12398-12401.
- [10] T.J. Garino, T.M. Nenoff, J.L. Krumhansl, D.X. Rademacher, J. Am. Ceram. Soc., 94 (2011) 2412-2419.
- [11] D.F. Sava, T.J. Garino, T.M. Nenoff, Industrial & Engineering Chemistry Research, 51 (2012) 614-620.
- [12] J.L. Krumhansl, T.M. Nenoff, Appl Geochem, 26 (2011) 57-64.
- [13] L.L. Burger, R.D. Scheele, PNNL-14860, (2004).
- [14] B.J. Riley, J. Chun, J.V. Ryan, J. Matyáš, X.S. Li, D.W. Matson, S.K. Sundaram, D. Strachan, J.V. Vienna, RSC Adv., 1 (2011) 1704-1715.
- [15] B.J. Riley, J. Chun, W. Um, W.C. Lepry, J. Matyáš, M.J. Olszta, K. Polychronopoulou, M. Kanatzidis, Environ. Sci. Technol., 47 (2013) 7540-7547.
- [16] C. Falaise, C. Volkring, J. Facqueur, T. Bousquet, L. Gasnot, T. Loiseau, Chem Commun., 49 (2013) 10320--10322.
- [17] S. Le Gallet, L. Campayo, E. Courtois, S. Hoffmann, Y. Grin, F. Bernard, F. Bart, J. of Nuclear Mat., 400 (2010) 251–256.
- [18] N. Torapava, C. Walther, Goldschmidt 2013 Conference, (2013) 2344.
- [19] D.S. Perera, B.D. Beff, D.J. Cassidy, R.L. Trautman, E.R. Vance, J. Aust. Ceram. Soc., 39 (2003) 81-87.
- [20] R.T. Jubin, Proceedings of the 17th DOE Nuclear Air Cleaning Conference, CONF-820833 (1982) 183–196.
- [21] J.R. Froines, P. Blanc, K. Hammond, D. Hattis, E. Loechler, R. Melnick, T. McKone, T. Slotkin, special Scientific Review Committee of the California Department of Pesticide Regulation, (2010).
- [22] J.G. Avila-Zárraga, R. Martínez, Synth. Commun., 31 (2001) 2177-2183.
- [23] S.D. Hellring, K.D. Schmitt, C.D. Chang, J. Chem. Soc., Chem. Commun., (1987) 1320-1322.
- [24] K.W. Chapman, P.J. Chupas, C.J. Kepert, J. Am. Chem. Soc., 127 (2005) 11232-11233.

- [25] P.J. Chupas, K.W. Chapman, G. Jennings, P.L. Lee, C.P. Grey, *J. Am. Chem. Soc.*, 129 (2007) 13822-13824.
- [26] R.T. Jubin, Airborne Waste Management Technology Applicable for Use in Reprocessing Plants for Control of Iodine and Other Off-Gas Constituents, Oak Ridge National Laboratory, Oak Ridge, TN, (1988).
- [27] T.M. Nenoff, J.L. Krumhansl, T.J. Garino, N.W. Ockwig, K. Holt-Larese, Waste Form Development and Testing for I₂, Sandia National Laboratories, AFCI-WAST-PMO-MI-DV-2009-000156(2008).
- [28] D. Haefner, N. Soelberg, INL/EXT-09-16837, (2009).
- [29] P.J. Chupas, X. Qiu, J.C. Hanson, P.L. Lee, C.P. Grey, S.J.L. Billinge, *J. Appl. Crystallogr.*, 36 (2003) 1342-1347.
- [30] P.J. Chupas, K.W. Chapman, P.L. Lee, *J. Appl. Crystallogr.*, 40 (2007) 463-470.
- [31] A.P. Hammersley, S.O. Svensson, M. Hanfland, A.N. Fitch, D. Hausermann, *High Pressure Res.*, 14 (1996) 235-248.
- [32] X. Qiu, J.W. Thompson, S.J.L. Billinge, *J. Appl. Crystallogr.*, 37 (2004) 678.
- [33] C.L. Farrow, P. Juhás, J.W. Liu, D. Bryndin, E.S. Božin, J. Bloch, T. Proffen, S.J.L. Billinge, *J. Phys.: Condens. Matter*, 19 (2007) 335219.
- [34] M. Wojdyr, *J. Appl. Crystallogr.*, 43 (2010) 1126-1128.
- [35] D. Haefner, T. Watson, INL/EXT-10-19657, (2010).
- [36] R.D. Scheele, L.L. Burger, C.L. Matsuzaki, PNL-4489, UC-70 (1983).
- [37] S. Hull, *Rep. Prog. Phys.*, 67 (2004) 1233-1314.
- [38] D.A. Keen, *J. Phys.: Condens. Matter*, 14 (2002) R819-R857.
- [39] P. Prager, *Acta Crystallogr., Sect. A: Found. Crystallogr.*, 30 (1974) 369-373.
- [40] V.M. Nield, D.A. Keen, W. Hayes, R.L. McGreevy, *Solid State Ionics*, 66 (1993) 247-258.
- [41] N.F. Uvarov, P. Vanek, M. Savinov, V. Zelezny, V. Studnicka, J. Petzelt, *Solid State Ionics*, 127 (2000) 253-267.
- [42] D.K. Murray, J.-W. Change, J.F. Haw, *J. Am. Chem. Soc.*, 115 (1993) 4732-4741.
- [43] D.K. Murray, T. Howard, P.W. Goguen, T.R. Kawietz, J.F. Haw, *J. Am. Chem. Soc.*, 116 (1994) 6354-6360.
- [44] Y. Jiang, M. Hunger, W. Wang, *J. Am. Chem. Soc.*, 128 (2006) 11679-11692.
- [45] Y.H. Ma, C. Mancel, *AIChE J.*, 18 (1972) 1148-1153.
- [46] W.M. Meier, *Z. Kristallogr.*, 115 (1961) 439-450.

Supplementary Data

Figure S1. Example of one μ -XRF spot analysis of $\text{CH}_3\text{I-Ag-MOR}$ as-received. A ratio of 1 Ag : 2.4 I is approximated in the sample.

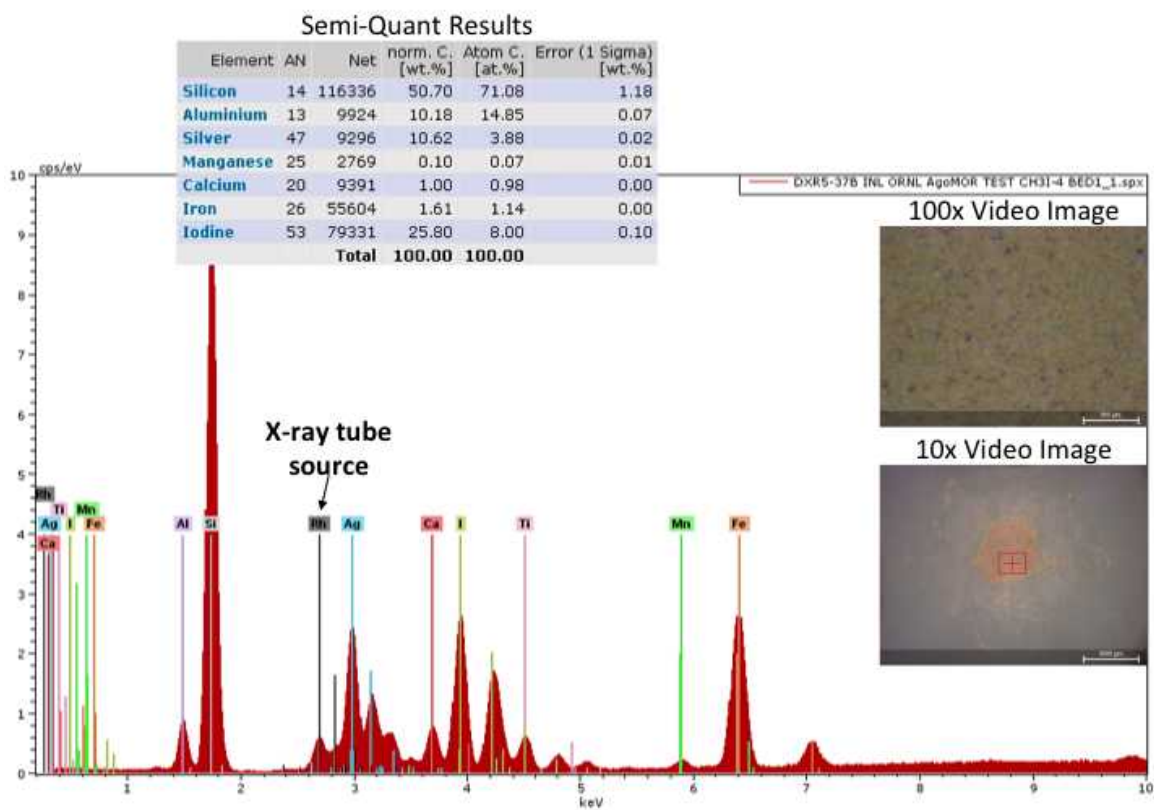


Figure S2. Example of one μ -XRF of $\text{CH}_3\text{I-Ag-MOR}$, after heating to 600°C . There is little to no change in the approximated ratio of Ag:I in the $\text{CH}_3\text{I-Ag-MOR}$ sample after heating to 600°C ($\sim 1 \text{ Ag} : 2.25 \text{ I}$).

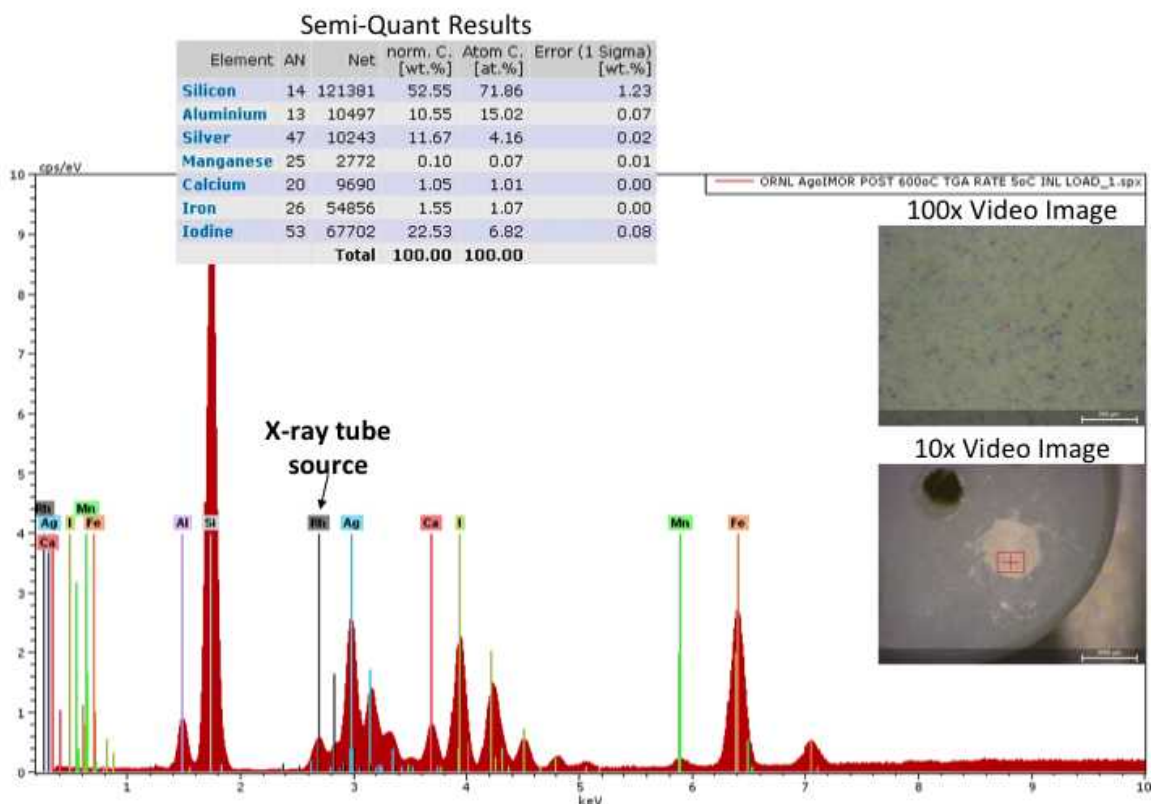


Figure S3. IR spectra of pristine (Ag° -MOR (natural), as received) as compared to the post exposure to CH_3I containing gas stream. There is no evidence of methyl groups permanently attached to the AgI -MOR structure.

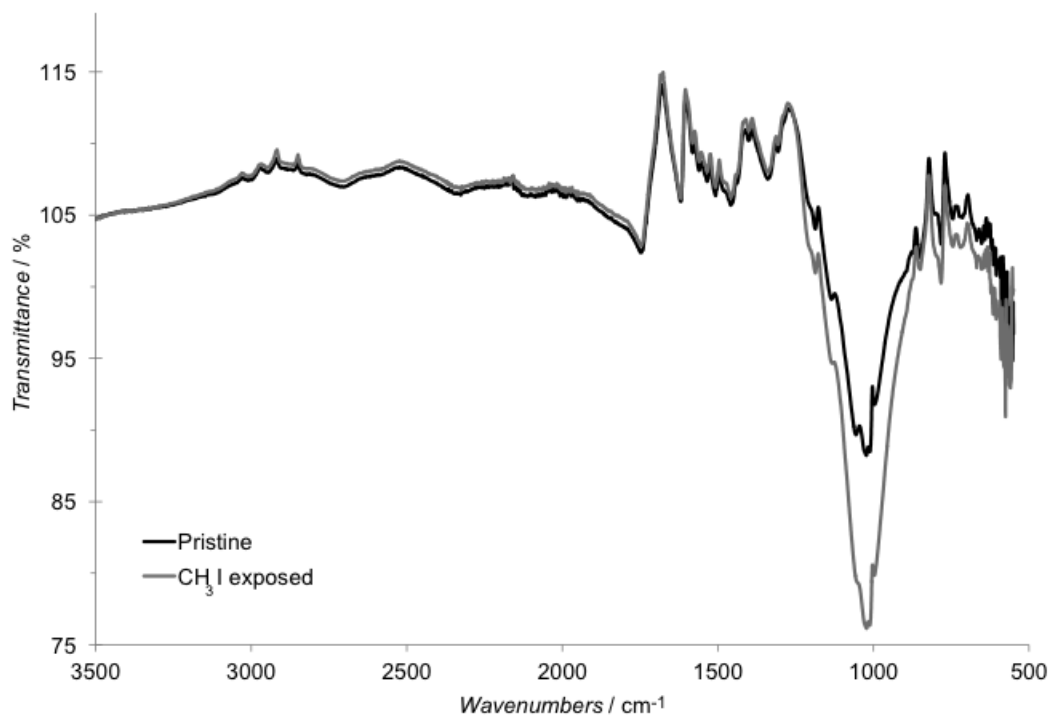


Figure S4. TGA-DSC-MS of AgI-MOR (natural, 8.7 wt% Iodine loaded) provided by S. Bruffey and R.T. Jubin, ORNL.

